# Jamie Fulford

Ph.D. Student in Computer Science, University of Illinois Urbana-Champaign
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## Education

**University of Illinois Urbana-Champaign** **August 2025 – Present**
*Ph.D. in Computer Science* *Champaign, Illinois*

- **Advisor:** Talia Ringer
- **Research interests:** programming languages, formal methods, interactive theorem proving, proof autoformalization

**University of Virginia** **August 2022 – May 2025**
*B.S. in Computer Science and Mathematics* *Charlottesville, Virginia*

- **GPA:** 3.9

## Publications

Swarn Priya, Frédéric Besson, Connor Sughrue, Tim Steenvoorden, **Jamie Fulford**, Freek Verbeek, and Binoy Ravindran. BeePL: Correct-by-compilation kernel extensions. *arXiv preprint* arXiv:2507.09883, 2025.

## Research Experience

**University of Illinois Urbana-Champaign** **August 2025 – Present**
*Graduate Research Assistant* *Champaign, Illinois*

- **DARPA expMath:** building tooling for proof autoformalization and proof decomposition in Agda, targeting homotopy type theory. The tooling decomposes a theorem statement into lemmas that are dispatched to LLM-based provers.
- **Proof translation between program logics:** designing languages with specific interpretable properties and implementing them in Rocq, so that the languages interoperate and proofs compile between them.

**Virginia Tech, Systems Software Research Group** **January 2025 – July 2025**
*Research Assistant (Proof Engineer)* *Remote*

- Developed type systems and compiler verification in Rocq for BeePL, a language for safe Linux eBPF kernel extensions.
- Extended the CompCert verified compiler with a new frontend language.

**University of Pennsylvania** **May 2024 – August 2024**
*NSF REU Research Fellow (Advisor: Steve Zdancewic)* *Philadelphia, Pennsylvania*

- Designed a lambda calculus for parallel and distributed computation and formalized it in Rocq, with machine-checked proofs of type safety and confluence.

## Projects

**Formalizing LLM Guardrails with AgentSpec in Lean 4** [paper] [code] **Fall 2025**

- Sole author. Executable Lean 4 semantics for AgentSpec, a DSL for runtime enforcement of LLM agent behavior, parameterized so the guarantees hold for any user-defined predicates.
- 42 machine-checked theorems with no `sorry`, including that safety is compositional in both directions, that the safety evaluator correctly implements the violation semantics, and that evaluation terminates.

**ProbaLean: A Probabilistic DSL in Lean 4** [paper] [code] **Spring 2026**

- With Trey Plante and Zirui Zhou (equal contribution). A discrete probabilistic programming language in Lean 4 with an executable operational semantics and a denotational semantics over s-finite measures, built on Mathlib.
- Proved that the interpretation is a monad morphism into the Giry monad, that independent sampling commutes, and that the denotational semantics is sound with respect to the operational semantics.

## Teaching Experience

Head Teaching Assistant, Computer Systems and Organization, University of Virginia (500+ students) **Fall 2023, Fall 2024**

## Honors & Awards

PLMW Scholarship, Programming Languages Mentoring Workshop at POPL 2025 **January 2025**

## Skills

**Proof assistants**: Agda, Rocq (Coq), Lean 4
**Verification**: Iris, CompCert, Mathlib
**Programming languages**: OCaml, Haskell, Rust, Python, C, C++, Java, JavaScript, SQL, Bash